

Chapter 3 Deterministic Modeling and Base Case Analysis

3.1 Steady-State Verification of the RELAP5 Model

Before we can even start looking at how the accident actually plays out, we first need to make sure that the RELAP5 model is giving us a stable and reasonable starting point. What I mean by this is that, if the model itself is not in a good steady condition before the transient event happens, then we would not be able to tell whether the changes we see later are actually coming from the ATWS-LOFW event or just from some numerical problem or noise in the code itself. So this step, the steady-state verification, it is actually quite important and we should not skip it.

When we look at the early portions of the simulation curves, what we can see is that all the major thermohydraulic variables they are staying more or less constant before anything happens in the transient. The pressurizer pressure for example, it starts at around 15.2 MPa which is what we expect for normal operating conditions. The steam generator pressure is beginning at roughly 5.2 MPa, and the reactor power is close to about 9.3 to 9.5 times 10 to the power of 8 watts. For the temperatures, the reactor average temperature it starts at approximately 616.7 K, and the peak cladding temperature is somewhere around 619.1 to 619.8 K depending on which channel we are looking at. The pressurizer level initially shows a value near 5.2 and then it quickly recovers up to around 9.8 to 10.0, which is actually a sign that the primary system control is already responding and working properly even in the very early part of the simulation.

Now there is one thing I should mention about the secondary side. The initial water inventory, it is not exactly the same in every steam generator channel. From the control-level plot we can read that SG level 160 starts near 4.8, SG level 260 is near 9.5, SG level 560 is near 6.0, and SG level 561 is around 5.5. At first someone might think this is an error or something wrong with the model, but actually it is not. What it reflects is the specific loop-by-loop conditions that were used in setting up the base-case calculation. The important thing here is not that all channels have exactly the same number, but rather that none of these channels show any strange or unphysical divergence or instability before the feedwater loss sequence begins. That is what matters for confirming the steady state.

Looking at the feedwater curves gives us even more confidence that the initial state is valid and correct. The main feedwater flow at the start of the simulation is about 260 kg/s, and the auxiliary feedwater flow is essentially at zero. This is exactly what we would expect to see during normal plant operation before any kind of loss-of-feedwater disturbance has occurred. So when we put all of these observations together, the stable initial values, the reasonable temperature and pressure readings, and the proper feedwater behavior, we can say with good confidence that the RELAP5 model is acceptable and suitable to serve as the deterministic reference condition for the ATWS-LOFW analysis that follows.

From a physical perspective what we should understand is that once the transient begins, the secondary heat sink gets suddenly degraded while the reactor core does not undergo an immediate conventional shutdown. This is what makes the scenario conservative, because basically the plant has to rely on its own thermal-hydraulic feedback mechanisms and whatever protective system actions are available, rather than depending on a prompt scram to shut things down quickly.

3.2 Conservative Assumptions and Boundary Conditions

The deterministic base case in this study it represents what we call an anticipated transient without scram that is combined with a loss of feedwater event. What makes this scenario conservative is that the plant is forced into a situation where it cannot rely on the normal reactor trip mechanism and at the same time it loses its primary method of removing heat through the secondary side. So the system has to manage itself using only the feedback effects and whatever protection systems are designed to activate automatically.

The most obvious and clearest boundary condition that we can see from the graphs is the loss of main feedwater. What happens is that the main feedwater flow it collapses very suddenly from about 260 kg/s all the way down to zero, and this happens very early in the transient period. On the other hand, the auxiliary feedwater does not come in immediately and when it does become available, it only provides a much smaller flow rate, approximately 10 to 12 kg/s. So what this means in practical terms is that the plant loses the vast majority of its normal secondary-side heat removal capability, and the only thing left afterward is this quite limited compensatory heat sink from the auxiliary system.

When we examine the valve-flow plots, they show us that the simulation does include functioning pressure-relief and protection responses on both the primary side and the secondary side of the plant. On the primary side, the pressurizer safety valve opens for a brief period with a peak flow reaching about 50 kg/s. On the secondary side, the steam generator safety valve opens much more sharply, showing a short peak that reaches near 260 kg/s, and then it closes back. After that, the steam generator release valve is the one that stays active for a longer period of time and it becomes one of the most important control elements during the later portion of the transient. This release valve peaks at near 100 kg/s initially and then later it goes into a cycling pattern where it repeatedly opens and closes between zero and approximately 45 to 50 kg/s.

I should note that the exact RELAP5 deck input values, things like the specific trip setpoints, control logic details, and actuation criteria, these are not directly visible from the figures alone and they should be reported separately in the methodology chapter or in an appendix. But even without seeing those exact input numbers, the behavior shown in the graphs is sufficient for us to understand what the effective boundary conditions are for this base case. Basically we have: loss of main feedwater, limited auxiliary feedwater recovery, secondary overpressure occurring, a brief primary relief event, and then a long-term pressure-control regime that is dominated by the cycling of the steam generator release valve.

3.3 Deterministic ATWS-LOFW Base-Case Results

To understand what happens in the deterministic response of the base case, we need to go through the thermal-hydraulic variables one by one in a logical sequence. The way the accident progresses is that first the feedwater loss removes the normal secondary heat sink from operation. After that, there is a rapid rise in pressure, followed by strong valve actions on both sides of the system. Then we see a significant reduction in the reactor power level, and there are temporary increases in both fuel and coolant temperatures. Eventually, the system moves into what I would describe as a later oscillatory regime, where the plant is no longer in its original steady state but it has transitioned into a controlled and lower-energy response condition.

Table 3.1 below presents the initial operating conditions of the system before any disturbance takes place. What it shows us is that the reactor is running under stable conditions that are near full power, with the pressures in both the pressurizer and the steam generator at their nominal values.

The temperatures for the reactor and the cladding they remain within the safe limits, which means there is no immediate risk of any damage at this starting point.

Also we can see that the normal main feedwater flow confirms the secondary-side heat removal system is working as expected, while the fact that auxiliary feedwater flow is at zero tells us there is no emergency demand at this stage yet. The pressurizer level behavior reflects normal control and inventory response. So overall, these initial values they give us a steady baseline that we can use as a reference point for comparing with what happens during the transient events that follow. These initial parameters are essential because without them we would not be able to clearly identify how much the system deviates from normal operation.

Table 3.1 Deterministic ATWS LOFW base case result

Variable	Graph-read initial value	Interpretation
Pressurizer pressure	15.2 MPa	Nominal primary-side pressurization
Steam generator pressure	5.2 MPa	Nominal secondary-side pressure
Reactor power	$9.3\text{--}9.5 \times 10^8 \text{ W}$	Near full-power initial condition
Reactor average temperature	616.7 K	Stable primary temperature before escalation
Peak cladding temperature	619.1–619.8 K	Cladding initially below any damage concern
Main feedwater flow	260 kg/s	Normal secondary-side heat sink available at start
Auxiliary feedwater flow	0 kg/s	No AFW demand before disturbance
Pressurizer level	5.2 then recovers to 9.8–10.0	Primary inventory/control response visible

3.3.1 AFW and MFW Behavior

The feedwater flow graph in Figure 3.1 is showing us one of the most important features of the entire transient, both in terms of what initiates the accident and also how the recovery attempt works. At the start of the simulation, the main feedwater which we call MFW is being maintained at approximately 260 kg/s. This value it tells us that the secondary-side heat removal conditions are

normal and the plant is operating as expected. This flow rate stays nearly constant only during the early stage of the simulation.

But then at around 200 seconds, something very significant happens. The MFW flow drops very abruptly all the way down to zero. This sudden drop is what clearly marks the onset of the loss of feedwater event that we are studying. Now what is interesting is that the auxiliary feedwater, or AFW, does not begin right away when the main feedwater is lost. Instead there is a delay period where the AFW remains at zero, and it only starts coming in after about a 30 second delay. When it does start, the flow increases to roughly 10 to 12 kg/s and then it stays at approximately that level for the remainder of the transient.

What this behavior is telling us is quite clear. The accident is characterized by a sudden and almost complete loss of the normal heat sink. The MFW goes from 260 kg/s to zero which is a very large drop. And the recovery through AFW at only 10 to 12 kg/s, while it does provide some cooling, it is really too small to compensate for the original MFW capacity. So even though some secondary-side cooling gets restored through the auxiliary system, the overall heat removal capability of the steam generator remains very much degraded throughout the entire transient.

Figure 3.1 Main feedwater and auxiliary feedwater

The shape of these curves is also worth paying attention to. The MFW reduction it is abrupt and once it drops it does not come back, so it is irreversible. The AFW recovery on the other hand is delayed and when it does come it stays far below the original feedwater level. This means that the system response in the later part of the transient is not governed by any restoration of normal feedwater conditions. Instead, what drives the later behavior is the combined effects of reduced heat removal, the depletion of inventory in the steam generators, the changing thermal-hydraulic conditions throughout the system, and the various protective valve actions that activate.

3.3.2 Steam Generator Release-Valve Flow Rate Behavior

The steam generator release-valve flow rate shown in Figure 3.2 highlights what I consider to be a very important pressure-control feature of the base-case transient. At the beginning of the simulation, the release-valve flow is at zero, and this makes sense because the steam generator pressure at that point is still below the threshold that would cause the valve to open. But after the

transient begins to develop, the valve opens quite rapidly and the discharge flow rises up sharply to reach a peak of about 100 kg/s.

This early high flow tells us that there is strong steam release happening from the secondary side as the pressure builds up. The pressure is building because the normal heat removal conditions have been degraded due to the loss of feedwater. After the flow reaches its maximum value, it gradually comes back down and falls to around 55 to 60 kg/s by approximately 700 seconds into the simulation. This decline shows us that the initial strong pressure relief is becoming weaker as the transient continues and the thermodynamic conditions inside the steam generator keep changing.

But here is what is interesting about this valve behavior. The valve does not just open once and then stay fully closed after that. Instead, from roughly 800 seconds onward, the Figure 3.2 shows a series of repeated narrow spikes in the flow. What is happening is that the flow repeatedly rises to nearly 45 to 50 kg/s and then returns back to zero, and this pattern repeats itself over and over. This cycling behavior is actually quite significant from a thermal-hydraulic point of view because it means the steam generator release valve is essentially acting as the main long-term pressure regulating mechanism. The fact that it keeps cycling like this in a repetitive way shows that the system has entered a kind of dynamic equilibrium state where the valve periodically opens to relieve pressure and then closes when the pressure drops enough.

Figure 3.2 Steam generator release-valve flow rate

3.3.3 Reactor Power Behavior

The reactor power response is one of the most dramatic changes we see in the entire transient sequence. What happens is that after the loss of feedwater event starts, the reactor power does not simply decrease in a smooth and gradual way. Instead, it drops very sharply and by a very large amount. The power falls from its initial level of about 9.3 to 9.5 times 10 to the power of 8 watts down to a much lower residual level, and this drop happens within just several hundred seconds of the transient beginning.

This rapid collapse of the reactor power is actually a very important safety feature of the system response. The reason it matters so much is that once the power drops to this low residual level, the core is no longer producing enough heat to continue driving the temperatures upward indefinitely.

In other words, the power suppression is what prevents the fuel cladding temperature from continuing to escalate without any bound. The thermal-hydraulic feedback mechanisms of the reactor, particularly the negative reactivity feedback effects, are what cause this power reduction even in the absence of a mechanical scram.

Looking at the power curve in more detail, we can see that after the initial sharp drop, the power does not go all the way to zero. Instead it settles at a low residual level and stays there. Some small oscillations may be visible in the later portion, but overall the power remains suppressed compared to its original full-power condition. This behavior is consistent with what we would expect from the physics of the reactor under these conditions, and it gives us confidence that the RELAP5 model is capturing the important feedback effects correctly.

3.3.4 Pressurizer Pressure Response

The pressurizer pressure response provides us with one of the clearest pictures of what is happening on the primary side of the plant during this transient. Initially the pressure is at around 15.2 MPa which is the normal operating pressure. But after the loss of feedwater begins, the pressure starts to rise because the heat that is normally being removed through the steam generators is no longer being taken away as effectively. The primary coolant heats up and expands, which pushes the pressure upward.

The pressure rises to a peak in the range of about 16.6 to 17.1 MPa. This is the period of greatest challenge for the primary system because the pressure is significantly above the normal operating value. At this point, the pressurizer safety valve opens briefly to relieve some of the excess pressure, with a peak flow of about 50 kg/s. After this initial peak, the pressure does not stay at its highest level. Instead it begins to come down as the reactor power suppresses and the thermal conditions start to change.

What is quite interesting about the pressurizer pressure behavior is what happens in the later part of the transient. The pressure first falls to approximately 13.8 to 13.9 MPa, which is actually below the initial operating pressure. Then it recovers and stabilizes in an oscillatory band near about 16 MPa. These oscillations in pressure correspond closely with the cycling behavior of the steam generator release valve that we discussed earlier. This connection makes sense physically because

the release valve cycling affects the secondary-side heat removal, which in turn affects the primary-side pressure through the heat transfer between the two sides.

3.3.5 Peak Cladding Temperature Behavior

The peak cladding temperature or PCT is perhaps one of the most safety-critical parameters we need to look at in any nuclear accident analysis, because if the cladding temperature goes too high, there is a risk of fuel damage. In our base case simulation, the PCT starts at approximately 619.1 to 619.8 K, which is well within the normal safe operating range.

After the transient begins and the feedwater is lost, the PCT rises because the heat removal from the core is degraded. The temperature goes up during the initial phase of the accident when the secondary heat sink is reduced but the auxiliary feedwater has not yet fully come into effect. However, and this is the encouraging part, the PCT does not continue to rise indefinitely. Instead, after reaching an elevated level during the early-to-intermediate period, it starts to come back down as the reactor power suppresses and some degree of secondary-side cooling is restored through the auxiliary feedwater.

In the later part of the transient, the PCT settles into a relatively bounded oscillatory pattern. The fact that the temperature remains bounded and does not show any runaway increase is very important from a safety perspective. It tells us that even under these challenging ATWS-LOFW conditions, the combination of reactor feedback, power suppression, and limited AFW recovery is sufficient to prevent the cladding temperature from reaching dangerous levels within the time frame of the simulation.

3.3.6 Steam Generator Pressure

The steam generator pressure response shows us what is happening on the secondary side from a pressure standpoint. At the start, the SG pressure is at about 5.2 MPa which is the nominal value. After the main feedwater is lost, the steam generator pressure rises quite rapidly because the steam that is being produced in the steam generators can no longer be condensed and removed by the normal feedwater flow. The steam builds up and the pressure increases.

The SG pressure reaches a peak of around 12.3 to 12.5 MPa, which is quite high compared to the initial value. This rapid overpressure is what triggers the opening of the steam generator safety valve, which we mentioned earlier discharges at a peak flow of about 260 kg/s. After this initial safety valve action relieves the worst of the overpressure, the SG pressure starts to come down and eventually enters an oscillatory pattern that corresponds to the release valve cycling behavior.

This pressure response on the secondary side is very important because it directly drives much of what we see on the primary side as well. The steam generator pressure rise is essentially the first major challenge that the system faces, and how the system handles this challenge through valve actions and heat transfer changes determines the subsequent evolution of the entire transient.

3.3.7 Steam Generator Level Response

The steam generator control-level plot in Figure 3.7 shows what I think is the strongest loop-to-loop asymmetry in the entire set of results from this base case. What I mean by this is that the different steam generator channels do not all behave in the same way, and the differences between them are quite significant and worth discussing in detail.

Starting with SG level 260, it begins near 9.5, then it rises to about 12.0 during the period between roughly 350 and 500 seconds, and after that it gradually declines. By the end of the plotted time interval, this channel is close to 6.0. SG level 560 starts near 6.0, rises slightly above 8.0 during the early phase of the transient, and then decreases continuously until it reaches essentially zero at about 850 to 900 seconds. SG level 561 starts near 5.5, climbs to around 7.5 to 7.8, and then remains comparatively elevated for much of the transient before gradually declining toward approximately 6.0 by the time we reach 2000 seconds. And finally SG level 160 begins near 4.8, falls rapidly after the feedwater-loss sequence develops, and reaches nearly zero by roughly 700 to 800 seconds.

These curves are showing us something very important, which is that the loss of feedwater does not affect all steam generators in exactly the same way or at the same rate. Two of the channels, especially SG level 160 and SG level 560, they experience near-complete depletion of their water inventory. The other two channels manage to retain a significant amount of inventory, although it is declining over time. This matters a lot because the secondary-side water inventory is what directly controls how much heat can continue to be removed from the primary system. So when the

depletion is uneven like this, it means that the different loops have uneven capability for extracting heat from the primary side.

These level results need to be interpreted together with the feedwater and valve-flow curves that we discussed earlier. Since the main feedwater drops from about 260 kg/s to zero and the auxiliary feedwater is only providing around 10 to 12 kg/s, there is simply not enough incoming water to fully restore the original secondary inventory. So the steam generator levels that we see represent one of the clearest signatures of the reduced heat sink condition in the ATWS-LOFW base case.

Figure 3.7 Steam generator control levels for the four modeled channels

3.3.8 Valve Actuation and Key Event Timeline

The sequence of how the different valves activate and when they activate provides us with what I think is the clearest operational timeline of how the entire transient progresses from beginning to end. Let me walk through it step by step so we can understand the full picture.

Right after the initiating event occurs, the main feedwater flow falls from about 260 kg/s down to zero. This is the moment that effectively marks the start of the loss-of-feedwater condition. Very shortly after that, the auxiliary feedwater becomes available, but as we discussed, it only comes in at a much lower flow rate of around 10 to 12 kg/s, which is nowhere near enough to replace the original secondary heat sink. As the steam generators lose their normal feedwater support, the pressure on the secondary side rises sharply. The steam generator pressure graph shows it going from roughly 5.2 MPa up to a peak near 12.3 to 12.5 MPa. This rapid overpressure triggers a short but high-magnitude opening of the steam generator safety valve, which reaches approximately 260 kg/s and then quickly returns to zero. So this safety valve acts as an early protective response rather than something that stays open for a long time.

On the primary side, the pressurizer pressure rises into the range of 16.6 to 17.1 MPa and this triggers a brief opening of the pressurizer safety valve. The flow from this valve peaks near 50 kg/s and then drops back to zero. The steam generator release valve however, it stays active for much longer than either of the safety valves. It climbs rapidly to nearly 100 kg/s, then gradually declines, and eventually enters into that repeated open-close cycling mode we talked about, with pulse amplitudes of approximately 45 to 50 kg/s.

When we view all the valve actions together alongside the power and temperature curves, the full picture of the accident progression becomes clear to us. The event starts with a sudden loss of the secondary heat sink, passes through an early overpressure phase, then protective valve operations occur on both sides of the system, and finally the system transitions into a lower-power oscillatory regime where the release valve plays the continuing role of regulating the secondary pressure.

Table 3.2 Valve actuation and key event timeline

Approximate time	Events	Graph value
0–150 s	Initial condition	Pressure, power, and temperature remain close to starting values
150–200 s	Main feedwater is lost	MFW drops abruptly from 260 kg/s to 0 kg/s
180–220 s	Auxiliary feedwater becomes available	AFW rises from 0 to 10–12 kg/s
200–450 s	Early overpressure develops	SG pressure rises toward 12.4 MPa, PZR pressure rises toward 17.1 MPa
200–260 s	Steam generator safety valve opens	Short SG safety-valve discharge peak 260 kg/s
240–300 s	Pressurizer safety valve opens briefly	Narrow PZR safety-valve peak 50 kg/s
250–800 s	Reactor power collapses and temperatures decline	Power drops by orders of magnitude; T avg and PCT trend downward after early peaks
800 s onward	Late oscillatory regime established	PZR pressure, T avg, PCT, and SG release valve show persistent cyclic behavior

3.4 Water Vapor Mass Fraction

From the deterministic base case results, we are able to identify which parameters are the ones that control the transient most strongly. Let me go through them one by one because understanding these controlling parameters is really crucial for the analysis.

The first and most obvious controlling parameter is the loss of main feedwater itself. Because the MFW flow drops from approximately 260 kg/s all the way to zero, the steam generators lose their normal ability to remove heat from the primary system. This is really the root cause of everything else that happens in the transient, and every other major response that we see in the graphs basically follows from this initial degradation of the secondary heat sink.

The second controlling parameter is the limited recovery that is provided by the auxiliary feedwater. Even though the AFW system does start up, its available flow of roughly 10 to 12 kg/s is simply too small to restore the original heat-removal capacity that was provided by the main feedwater. This limited makeup is the reason why some of the steam generator levels fall nearly to zero during the transient, and it also explains why the system cannot simply return to its original operating point after the transient begins. The AFW provides some relief, yes, but it is not enough to fully compensate.

The third controlling parameter is the secondary-side pressure relief mechanism. The steam generator safety valve handles the immediate overpressure situation, but it only does so briefly. After that initial relief, the steam generator release valve takes over and becomes the principal long-term regulating element for the secondary pressure. The repeated pulsing behavior of this valve that we observed corresponds very closely with the late oscillations that we see in both pressure and temperature, which shows that this valve has a strong and direct influence on the final dynamic balance of the system.

Fourth, we have the primary-side pressure evolution as a controlling parameter. The pressurizer pressure first rises to about 17.1 MPa, then later falls to approximately 13.8 to 13.9 MPa, and after that it stabilizes in an oscillatory band near 16 MPa. This pressure history reflects the combined effect of coolant expansion, relief actions, power suppression, and the changing heat transfer conditions between the primary and secondary sides. It also has influence on the saturation margins and system inventory behavior in the primary loop.

The fifth controlling parameter is the rapid suppression of reactor power. Because the power drops from nearly 10 to the power of 9 watts down to a low residual level within several hundred seconds, the core is no longer driving continuous temperature escalation. The fact that the cladding temperature stays bounded is directly linked to this power collapse. And finally, the unequal distribution of steam generator inventory among the four channels is also a controlling factor. This matters because it determines how unevenly the remaining heat sink is shared among the different loops.

Table 3.3 Water Vapor Mass Fraction

Controlling parameter	Graph value	Why it matters
Main feedwater loss	MFW 260 kg/s → 0 kg/s	Initiates the loss of the normal secondary heat sink
Limited AFW recovery	AFW stabilizes only at 10–12 kg/s	Insufficient to restore original heat removal
Secondary pressure relief	SG safety valve peak 260 kg/s, release valve peaks 100 kg/s then pulses	Controls early and late secondary pressure evolution
Primary pressure response	PZR pressure 15.2 → 17.1 → 13.8 → oscillatory 16 MPa	Reflects overall primary-side thermohydraulic balance
Power suppression	Reactor power falls from 9.5×10^8 W to low residual level	Limits continuing fuel and cladding heating
SG inventory distribution	Two SG levels approach zero, two remain partially filled	Determines the remaining loop-wise heat sink capability

3.5 Chapter Summary

So to summarize what we found in this chapter, the RELAP5 deterministic base-case results show us a coherent and physically interpretable response to the combined ATWS-LOFW scenario. The model begins from a stable initial state, with the primary pressure near 15.2 MPa, the secondary pressure near 5.2 MPa, the reactor power close to full power, the average temperature near 616.7 K, and the peak cladding temperature near 619.1 to 619.8 K.

The transient progression that we observed follows a clear and logical sequence. First the main feedwater is lost and the secondary heat sink is degraded. Then the pressures rise on both sides of the system and the safety valves open to provide initial protection. The reactor power drops dramatically due to thermal-hydraulic feedback. The temperatures show an initial excursion and then come back down. And finally the system enters a long-term oscillatory regime where the steam generator release valve plays the dominant role in controlling the secondary pressure.

What is most important from a safety perspective is that all the key parameters, the pressurizer pressure, the peak cladding temperature, the steam generator pressures and levels, they all remain within bounded ranges throughout the transient. There is no runaway or unbounded escalation in any of these variables. This gives us confidence that the deterministic base case represents a challenging but manageable scenario, and it provides a solid foundation for the probabilistic and dynamic event tree analyses that follow in Chapter 4.

Chapter 4 MCDET-Based Analysis of ATWS Loss of Feedwater

4.1 Introduction

In this chapter we move beyond the single deterministic calculation that was presented in Chapter 3 and we start using the Monte Carlo and Dynamic Event Tree methods to understand how uncertainty and different branch paths affect the accident response. The reason we need to do this is that a single deterministic run, while it is useful for understanding the basic physics and the sequence of events, it does not tell us about the range of possible outcomes that could happen when the input conditions are uncertain or when different system actions occur at different times.

The MCDET approach that we use in this chapter combines two important ideas. The first is Monte Carlo sampling, where we run the RELAP5 simulation many times with slightly different input parameters to see how the results vary. The second is the Dynamic Event Tree analysis, where we systematically explore different branch paths that the accident can take depending on which systems activate and in what combination. Together these two methods give us a much more complete and much more realistic picture of the accident than what any single deterministic calculation could provide.

4.2 Monte Carlo Simulation Setup

The Monte Carlo simulation involves running the RELAP5 model multiple times, and each time the uncertain input parameters are sampled from their defined distributions. The key uncertain parameters that we considered include things related to the timing of feedwater loss, the delay in auxiliary feedwater activation, and other relevant system parameters that could vary in a real plant situation. By sampling these parameters many times and running the simulation for each sample, we generate a collection of transient trajectories that we can analyze statistically.

The advantage of doing it this way is that instead of having just one answer for each safety parameter, we get a whole distribution of answers. This distribution tells us not just what the most likely outcome is, but also what the range of possible outcomes looks like and how likely the more extreme outcomes are. This kind of information is much more valuable for safety assessment than a single deterministic number.

4.3 Monte Carlo Results and Analysis

4.3.1 Impact of Feedwater Parameters

When we look at the Monte Carlo results for the feedwater behavior, what we see is that all the trajectories for the main feedwater flow show a similar overall pattern. The MFW flow starts at the normal operating level and then drops abruptly to zero, just like in the deterministic base case. However, the exact timing of when this drop happens varies slightly from one realization to another due to the uncertainty in the sampled parameters. The bundle of trajectories is relatively tight, which tells us that the MFW collapse behavior is quite robust and not strongly affected by the parameter uncertainties we considered.

The auxiliary feedwater response shows what we can call a complementary behavior to the MFW. The AFW flow rate starts from zero and then rises sharply to a nearly constant plateau once the auxiliary system becomes established and operational. The bundle of trajectories shows only a small spread in the final stabilized AFW delivery rate, which means the main uncertainty is concentrated more in the actuation timing rather than in the eventual flow magnitude. This result is actually quite significant from a safety standpoint because what it tells us is that the short interval between when the MFW collapses and when the AFW becomes available, this interval is what governs how severe the intermediate thermal-hydraulic excursion will be. In other words, the early transient response is controlled less by how much AFW eventually delivers, and more by how quickly the auxiliary pathway becomes available after the normal feedwater is lost.

From a safety interpretation standpoint, these feedwater plots are demonstrating to us that the accident evolution is strongly conditioned by the question of secondary-side heat sink replacement. The prompt reduction of MFW is what creates the initial challenge to the plant, and the establishment of AFW is what creates the mechanism for later stabilization. Therefore, what we can conclude is that any increase in the AFW delay time or any decrease in the delivered AFW capacity would be expected to make the pressure excursion worse and increase the thermal burden on both the core and the primary system. This observation gives us a clear basis for selecting AFW-related variables as important branching parameters in the later DET analysis.

4.3.2 Impact of Peak Cladding Temperature

Peak cladding temperature is one of the most important indicators that we look at when assessing fuel integrity and short-term thermal safety. As shown in Figure 4.2, what we can see is that the PCT does not increase in a monotonic fashion throughout the whole transient. Instead, what happens is that it first rises from its initial level to an early local maximum, then it decreases down to a minimum value during the intermediate recovery stage, and finally it settles into a relatively narrow oscillatory band. The shape of this response is telling us that the cladding temperature is highly sensitive to the balance between two competing effects: the persistent heat source that exists because of the ATWS conditions where the reactor has not scrammed, and the gradual restoration of secondary-side heat removal as the AFW comes into operation.

Figure 4.2 Peak cladding temperature during the ATWS-LOFW transient

The spread among the different Monte Carlo trajectories is relatively limited, and this is actually an important finding. What it suggests is that the present set of uncertainties perturbs the timing of the thermal response more strongly than its ultimate amplitude. What I mean by this is that the different runs reach their peak temperatures at slightly different times, but the actual peak values are not dramatically different from each other. The early local peak in PCT corresponds to the period immediately following the degradation of the feedwater function, when the system heat sink has been reduced but the compensating AFW effect has not yet fully developed and taken hold. Once the AFW is established and the primary-to-secondary heat transfer condition improves, the PCT decreases noticeably, and this indicates a temporary recovery of the thermal margin.

The subsequent stabilization of PCT in a compact band is what I would call an encouraging result from a safety perspective. It implies that none of the sampled realizations, not a single one, produces a runaway increase in cladding temperature within the duration that we analyzed. Nevertheless, the early PCT excursion remains a safety-relevant feature that we should not ignore. It identifies the specific interval in which uncertainty in feedwater behavior has the greatest thermal consequence, and therefore it marks the period that deserves the greatest attention when we move to the sequence-based MCDET treatment.

4.3.3 Impact of Pressurizer Pressure

The pressurizer pressure response that is shown in Figure 4.3 reflects what is essentially the changing balance between several competing physical effects: the core heat generation, the coolant thermal expansion, and the effectiveness of heat removal to the secondary side. What we observe is an initial increase in pressure shortly after the start of the transient, followed by a multi-stage decline toward a minimum value, and then a recovery to a later quasi-steady oscillatory band. This non-monotonic profile, meaning it goes up then down then up again, is typical of what we see in a strongly coupled transient where the system first experiences a degradation in heat sink capacity and then re-adjusts itself once the auxiliary removal path becomes active and functional.

Figure 4.3 Pressurizer pressure during the ATWS-LOFW transient

The first pressure peak represents the period of greatest challenge for the primary system. Because the feedwater disturbance acts on the secondary side while the reactor heat generation persists under ATWS conditions, the primary system responds through pressurization. The Monte Carlo spread remains moderate across the different realizations, and this indicates that while the timing of the pressure excursion is somewhat variable from run to run, the general amplitude and the overall pattern are preserved across the entire sample set. In this sense, we can say that the pressurizer pressure behaves as what I would describe as a robust macroscopic indicator of how the transient is evolving.

A pronounced reduction in pressure occurs during the intermediate stage before the variable recovers toward its long-term band. This suggests that the system undergoes what might be called a temporary overcorrection or relief phase as the secondary-side cooling condition changes and the primary inventory and thermal state get redistributed. For the purpose of this thesis interpretation, the most important point to take away is that the pressurizer pressure clearly responds to uncertainty in feedwater behavior, and therefore it serves as a suitable response variable for both probabilistic ranking and later DET branching logic.

4.3.4 Impact of Steam Generator Pressure

Figure 4.4 presents the Monte Carlo response of the steam generator pressure, which is a direct measure of the thermodynamic condition on the secondary side of the plant. What we see is that the

pressure rises rapidly during the early part of the transient and reaches a distinct maximum before it gradually decreases down to a lower, nearly constant oscillatory band. This pattern is physically consistent with what we would expect from the loss of normal feedwater: once the secondary-side water replenishment is interrupted, the steam generator experiences a temporary deterioration in its heat removal conditions, and this deterioration manifests itself as a rise in pressure.

Figure 4.4 Steam generator pressure during the ATWS-LOFW transient

As with the other key variables that we discussed, the uncertainty-induced dispersion in the Monte Carlo results is concentrated mainly around the early and intermediate stages of the transient. The peak in steam generator pressure occurs during the period when the normal feedwater path has been lost and the AFW path is still in the process of transitioning to its effective operating condition. Once the AFW is fully established and working, the steam generator pressure falls and remains clustered within a comparatively narrow range. This tells us that the secondary side has reached what we can consider a new operational balance under the accident conditions.

What I find particularly important about these results is the close correspondence between the steam generator pressure trend and the primary-side indicators. The steam generator pressure rise precedes or accompanies the period of highest challenge in both pressurizer pressure and PCT. This demonstrates to us that the secondary-side performance is really the dominant driver of the observed transient behavior. The steam generator pressure can therefore be interpreted not merely as just an output variable, but also as what I would call a compact indicator of how adequate the replacement heat sink is. This justifies why we use it in the uncertainty interpretation of the Monte Carlo stage and, later on, in the sequence discrimination of the MCDET framework.

4.4 Dynamic Event Tree (DET) Analysis

Now we come to the Dynamic Event Tree part of the analysis, which is where things get really interesting from a safety analysis perspective. The DET results demonstrate to us that the ATWS loss-of-feedwater transient does not evolve along just a single deterministic path. Instead, what happens is that the system response separates into sixteen different branches, and each one of these branches represents a different combination of branch events that can occur during the accident.

When we look at the plotted responses for these branches, we can see that they diverge from each other in terms of several important parameters: the secondary-side inventory depletion, the pressurizer response, the peak cladding temperature, and the availability or timing of various mitigation functions. As a result of this divergence, the DET framework proves to be very useful for identifying not only the average behavior of the accident, but also which branches lead to stable outcomes, which ones lead to unstable outcomes, and which ones are rapidly terminated.

4.5 DET Methodology for ATWS Loss of Feedwater

In this study, the DET methodology is represented by sixteen branch combinations that are identified in the figures. These branches are: LOFW, LOFW+SP, LOFW+RV, LOFW+AFW, LOFW+SV, LOFW+RV+AFW, LOFW+RV+SV, LOFW+RV+SP, LOFW+SP+SV, LOFW+SP+AFW, LOFW+AFW+SV, LOFW+RV+SV+AFW, LOFW+RV+AFW+SP, LOFW+SP+AFW+SV, LOFW+RV+SP+SV, and LOFW+ALL. These labels they indicate the specific branch configuration that is used in the DET model. Each label tells us which combination of system actions or events is occurring in that particular branch.

From the graph set, we can interpret these branches in terms of four general response classes. The first class is what we call the rapidly terminated branch, where the transient ends relatively quickly. The second class is the highly stressed branch, which shows strong pressure escalation. The third class consists of intermediate branches that terminate around 1150 to 1260 seconds. And the fourth class includes longer-lasting branches that remain active until the end of the simulation window.

Table 4.1 DET Branch Interpretations

Sequence	Interpretation in the DET context
LOFW	Baseline accident branch used as the main reference for comparison
LOFW+SP	Modified-pressure branch that remains close to the baseline response
LOFW+RV	Long-duration stable branch that survives to the end of the simulation window
LOFW+AFW	High-stress branch with strong pressure escalation and clear deviation from stable group
LOFW+SV	Stable branch with bounded late-time behavior and no severe excursion

LOFW+RV+AFW	Moderately stable combined branch with more controlled response than highly stressed cases
LOFW+RV+SV	Intermediate branch showing partial mitigation but shorter persistence
LOFW+RV+SP	Long-duration combined branch that remains within the stable late-time cluster
LOFW+SP+SV	Stable combined branch with bounded pressure and temperature behavior
LOFW+SP+AFW	Partially stabilized branch that follows the main group but terminates earlier
LOFW+AFW+SV	Higher-stress combined branch with sustained late-time elevation
LOFW+RV+SV+AFW	Multi-action branch; intermediate severity
LOFW+RV+AFW+SP	Multi-action branch; controlled but elevated
LOFW+SP+AFW+SV	Late escalation branch; among the most unfavorable outcomes
LOFW+RV+SP+SV	Stable multi-action branch persisting through late time
LOFW+ALL	Full combination branch with all actions present

4.6 Representative DET Branch Analysis

To better understand how these different branches actually behave, let me go through some of the representative ones and discuss what their pressurizer pressure and peak cladding temperature responses look like. This will help us see the differences between the branches more clearly.

4.6.1 LOFW (Baseline Branch)

The LOFW branch serves as the baseline reference for all comparisons. In this branch, the pressurizer pressure follows the pattern we described in Chapter 3, rising to the initial peak and then settling into the oscillatory regime. The peak cladding temperature shows a similar bounded behavior. This branch belongs to what we classify as the bounded response group, meaning the late-time behavior is relatively smooth and both pressure and temperature end near the lower envelope of all the branches.

4.6.2 LOFW+SP (Spray Branch)

When spray is added to the baseline LOFW scenario, the response remains quite close to the baseline case. The pressurizer pressure shows very similar behavior, and the peak cladding temperature also stays within a similar bounded range. This tells us that the spray action by itself

does not dramatically change the accident trajectory. This branch also belongs to the bounded response group.

4.6.3 LOFW+AFW (Auxiliary Feedwater Branch)

This branch is one of the more interesting ones because adding auxiliary feedwater as a branch event actually leads to a high-stress response. The pressurizer pressure enters a broader oscillatory region with higher amplitudes, and the peak cladding temperature continues to rise into a higher range during the later part of the transient. This might seem counterintuitive at first, because one would think that having auxiliary feedwater available would be helpful. But what the DET analysis is showing us is that the specific timing and coupling of the AFW with the other system responses can lead to a different and potentially more challenging trajectory than the baseline case.

4.6.4 LOFW+RV (Relief Valve Branch)

The relief valve branch shows a long-duration stable response that survives all the way to the end of the simulation window. The pressure and temperature remain within bounded ranges throughout. This branch demonstrates that under this particular combination of conditions, the system reaches a sustainable operating state despite the challenging ATWS-LOFW conditions.

4.6.5 LOFW+AFW+SV Branch

This combined branch represents one of the more severe outcomes in the DET analysis. The pressurizer pressure remains high during the later stage and then rises sharply near the end of the transient, reaching values in the highest region visible in the pressure plots. The peak cladding temperature follows the same severity pattern and ends up in the highest thermal band among all the branches. This branch belongs to what we classify as the late escalation group, which represents the most unfavorable response class in our analysis.

4.6.6 LOFW+SP+AFW+SV Branch

This branch forms what is clearly the most severe late escalation case in our representative set of results. The pressure curve remains high during the later stage and then rises sharply near the end of the transient, reaching the highest region visible in all the pressure plots across all branches. The peak cladding temperature follows the same overall severity pattern and ends in the highest thermal

band. This confirms that the coupled accident evolution can indeed lead to distinctly more severe outcomes under certain specific branch combinations.

4.7 Comparative Discussion of Branch Severity

When we consider all the branches together and compare their behavior, we can group them into three broad response classes. This grouping is actually very useful because it helps us turn a complex set of graphs into a coherent accident narrative that we can understand and interpret.

The first class is what we call the bounded response group. This group is represented mainly by the LOFW and LOFW+SP branches. These branches show relatively smooth late-time behavior, with the pressure ending near the lower envelope of the entire set and the peak cladding temperature remaining close to the mildest thermal response. These are essentially the least concerning branches from a safety perspective.

The second class is the elevated oscillatory group. This includes branches like LOFW+AFW, LOFW+RV+AFW, LOFW+SP+AFW, and LOFW+RV+SP+AFW. These branches remain above the bounded group in both pressure and temperature, and they show sustained oscillatory behavior during the later stage of the transient. They are more challenging than the bounded group but they do not show the sharp escalation that we see in the worst cases.

The third class is the late escalation group, represented by LOFW+AFW+SV and LOFW+SP+AFW+SV. In these branches, both the pressure and the temperature rise into the most severe end-of-transient region visible in the current graph set. These are the branches that represent the greatest safety concern.

This classification is useful because it demonstrates something very important about what the MCDET coupling does. The method does not simply generate several mathematical branches for the same event and leave it at that. What it actually shows is that the exact same initiating accident can evolve toward very different levels of severity depending on which coupled branch path the system follows. Some branches remain comparatively mild, some settle into an elevated but still bounded response, and some progress toward a clearly more severe late-time condition. This is the main value of the coupling process presented in this chapter, because it converts the methodology that was developed earlier in the thesis into an interpretable set of branch-wise safety results.

The graphs that we presented show that the MCDET coupling process produces a meaningful branch-wise separation of the ATWS loss-of-feedwater transient when we interpret it through the lens of pressurizer pressure and peak cladding temperature. All branches share a common early stage behavior, which makes sense because they all start from the same initial conditions. But they diverge clearly in the later part of the accident as the different system actions take effect. The result is a structured set of bounded, elevated, and severe response paths that can be interpreted directly in terms of safety significance.

This chapter provides the important link between the methodology that was described in Chapter 2 and the actual safety-relevant results. It demonstrates that the combined RELAP5 Monte Carlo Dynamic Event Tree framework can indeed distinguish among different accident developments and show how those developments affect the two most important output variables that were selected for the present MCDET analysis.

4.8 Monte Carlo Mean Trajectories and Probability Distributions for PCT and Pressurizer Pressure

4.8.1 Introduction

In this section we present a more detailed statistical analysis of two key safety parameters from the Monte Carlo simulations. The first parameter is the peak cladding temperature or PCT, which describes the thermal response of the fuel. The second parameter is the pressurizer pressure, which describes the primary-side pressure response. For each of these parameters, we look at two things: the mean trajectory over time, which shows us the average physical evolution of the transient, and the probability distribution of the peak value, which tells us about the uncertainty in the maximum value reached.

4.8.2 Mean Monte Carlo Trajectory of Peak Cladding Temperature

The mean PCT trajectory shows that the thermal response of the core is strongly dependent on time and can be divided into several characteristic stages. At the beginning of the transient, the mean value remains close to about 619 to 620, which indicates that the system is still near its initial condition and nothing dramatic has happened yet.

Then a sharp temperature increase appears at approximately 200 seconds, and the family of Monte Carlo trajectories rises rapidly to the first elevated region. In this region, the mean response reaches roughly 628 to 629 and some individual realizations extend slightly above 630. This part of the curve represents the period in which the loss of feedwater weakens the secondary-side heat removal and temporarily intensifies the thermal challenge to the fuel cladding. It is during this window that the cladding faces its greatest thermal stress.

Figure 4.20 Mean Monte Carlo trajectory of peak cladding temperature during the ATWS-LOFW transient

After the early peak region, the mean trajectory does not collapse immediately to a final steady state. Instead, what happens is that it passes through a short intermediate shoulder and then decreases continuously until it reaches a clear minimum at around 780 to 800 seconds, where the mean value drops to nearly 611. This deep trough indicates that the plant response and heat-removal mechanisms become more effective after the initial thermal excursion. The subsequent recovery from about 800 seconds to 1000 seconds is smooth and well organized, and after about 1000 seconds the trajectories settle into a relatively narrow oscillatory band near 621.5 to 622.2.

From a safety interpretation perspective, this graph shows that the most severe thermal challenge occurs in the early-to-intermediate stage of the accident rather than in the long-term part of the transient. This is an important finding because it tells us where to focus our attention when assessing the safety margins.

What is also noteworthy is that the uncertainty in the sampled input parameters does change the detailed thermal path, but it does not alter the overall pattern of the response. All of the realizations follow the same basic sequence: an initial rise, a strong decrease, and then a later stabilization. This consistency across the different Monte Carlo runs is useful because it demonstrates both the dominant physical trend of the accident and the comparative robustness of the thermal response under the uncertainty sampling conditions.

4.8.3 Probability Distribution of Peak Cladding Temperature

The probability histogram of the peak PCT values complements the previous time-history figure by focusing specifically on only the maximum temperature that is reached in each Monte Carlo

realization. What we can see from this histogram is that the plotted peak values lie within a relatively narrow interval, approximately from 629.2 to 630.6, and there is a dashed vertical line that indicates a mean peak value of 629.9353.

The fact that this range is narrow is actually an important finding. It shows us that, although the transient contains noticeable uncertainty in various input parameters, the maximum thermal response remains clustered within a limited band and does not show wide divergence. The distribution is not concentrated at a single point, which means the peaks are spread across several neighboring bins. This implies that the sampled uncertainties do influence the exact maximum value reached by each run, so the uncertainty is real and meaningful. However, the spread is still small when we compare it with the full time-history change shown in the mean trajectory plot.

Figure 4.21 Probability distribution of peak cladding temperature obtained from Monte Carlo simulations

In other words, the simulation ensemble produces some variation in how high the peak gets, but there is not a large divergence in the thermal severity of the accident from one run to another. From the viewpoint of safety characteristic analysis, this figure should be interpreted as the probabilistic description of the thermal safety margin. It shows us that the peak cladding temperature is governed by uncertainty, but that this uncertainty is bounded and statistically well behaved. This strengthens the argument that the ATWS-LOFW thermal response is sensitive to input variation but remains within controllable and safe limits.

4.8.4 Mean Monte Carlo Trajectory of Pressurizer Pressure

The mean pressurizer pressure trajectory exhibits a transient pattern that closely parallels the thermal behavior we just discussed, but expressed in pressure units and with clear engineering relevance to primary-system safety. At the start of the simulation, the pressure decreases quickly from about 15.6 MPa to nearly 15.2 MPa, which shows an initial adjustment from the starting condition.

Then a sharp pressure rise occurs near 200 seconds, and the mean curve moves into a high-pressure region around 16.6 to 16.8 MPa, while some individual realizations rise slightly above 17.0 MPa. This phase corresponds to what we identified as the strongest primary-side challenge after the loss

of feedwater. The reduced heat removal on the secondary side drives the reactor coolant system toward higher pressure because the heat has nowhere to go efficiently.

Following this early high-pressure stage, the pressure does not remain at its maximum level. The mean trajectory passes through a short intermediate period and then decreases to a clear minimum, where the mean pressure drops significantly below the initial level. This dip represents the overcorrection phase we mentioned earlier. After this minimum, the pressure gradually recovers and eventually enters a quasi-steady oscillatory band in the later part of the transient.

The Monte Carlo spread in the pressure trajectories is moderate, which confirms what we observed with the PCT results. The timing of the pressure excursion varies somewhat from run to run, but the general shape and amplitude of the response are preserved across all the sampled realizations. This makes the pressurizer pressure a reliable and robust indicator for the MCDET safety analysis.

4.8.5 Probability Distribution of Pressurizer Pressure

The probability histogram of the peak pressurizer pressure values from the Monte Carlo runs complements the time-history trajectory by showing us the statistical distribution of the maximum pressure reached in each realization. Similar to what we saw with the PCT distribution, the peak pressure values are concentrated within a relatively narrow band, indicating that the uncertainty does not cause wild variations in the maximum primary pressure.

The distribution shape tells us that while individual runs reach slightly different peak pressures due to the parameter uncertainties, the overall range of variation is bounded and manageable from a safety perspective. This is consistent with the finding from the PCT analysis that the ATWS-LOFW transient, while sensitive to input uncertainties, does not produce extreme outlier responses under the conditions we studied.

4.9 Chapter Summary

To bring together everything that we have discussed in this chapter, the MCDET-based analysis has shown us several important things about the ATWS loss-of-feedwater transient that we could not have learned from the deterministic analysis alone.

First, the Monte Carlo simulation results show that the key safety parameters including peak cladding temperature and pressurizer pressure are sensitive to uncertainty in the input parameters, but the uncertainty effects are bounded and well behaved. The mean trajectories follow clear and physically interpretable patterns, and the probability distributions of the peak values are concentrated in narrow ranges. This gives us confidence that the ATWS-LOFW response, while variable, does not produce extreme or runaway outcomes under the conditions we analyzed.

Second, the Dynamic Event Tree analysis reveals that the accident can develop into multiple distinct branches with different levels of severity. We identified three main response classes: bounded branches, elevated oscillatory branches, and late escalation branches. This classification shows that the same initiating event can lead to very different outcomes depending on which combination of system actions occurs during the transient.

Third, the MCDET coupling framework successfully combines the Monte Carlo uncertainty treatment with the systematic branch exploration of the DET method. This combined approach provides a much richer and more informative safety characterization than either method could provide on its own. It demonstrates that the RELAP5-based MCDET framework is a viable and useful tool for analyzing complex nuclear safety scenarios like the ATWS-LOFW transient in pressurized water reactors.

The results from this chapter provide the direct safety-relevant findings of the thesis and demonstrate the practical value of the integrated MCDET methodology that was described in Chapter 2 and the deterministic foundation that was established in Chapter 3.